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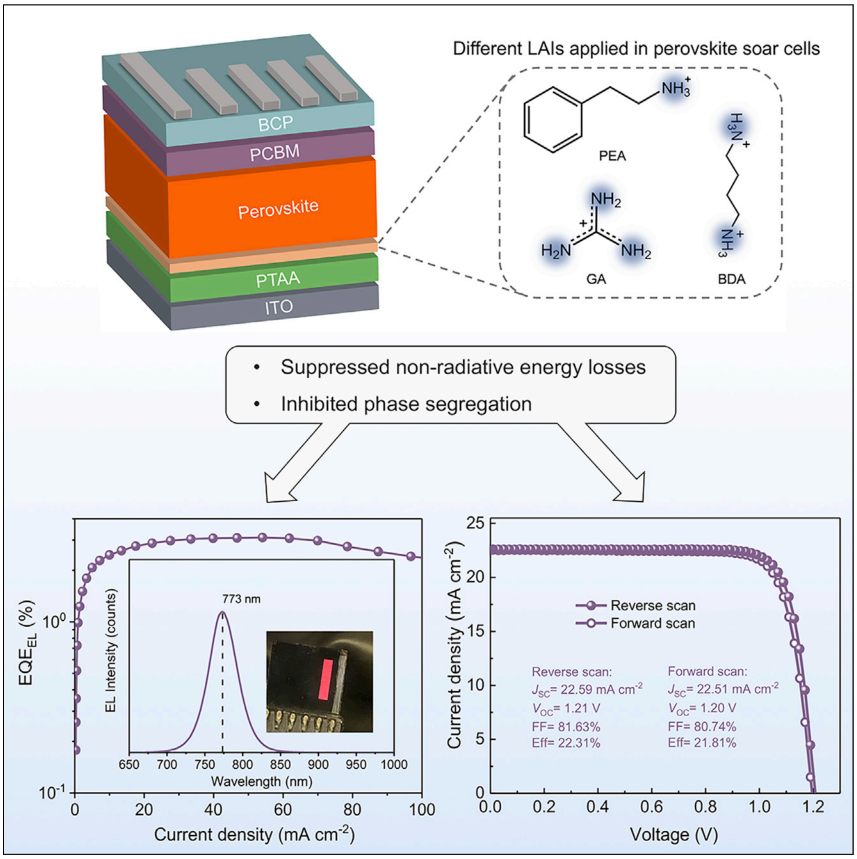

The non-radiative energy losses at both top and bottom interfaces of perovskite were simultaneously suppressed by introducing LAIs between the hole transport layer and perovskite layer. More importantly, the LAIs have also effectively inhibited the phase segregation on the perovskite surface, enabling homogeneous surface properties. As a result, a champion efficiency of over 22% was realized for p-i-n structured PVSCs, which is among the highest efficiencies reported for p-i-n structured PVSCs.

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HIGHLIGHTS

Large alkylammonium interlayers (LAIs) reduce the interfacial non-radiative losses

LAIs suppress the phase segregation at the top surface of perovskite layer

The LAI-based device show improved efficiency from 20.09% (Control) to 22.31%

Wu et al., Joule 4, 1248–1262
June 17, 2020 © 2020 Elsevier Inc.
https://doi.org/10.1016/j.joule.2020.04.001

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Modulation of Defects and Interfaces through Alkylammonium Interlayer for Efficient Inverted Perovskite Solar Cells

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SUMMARY

Perovskite solar cells (PVSCs) with p-i-n configuration bear great potential for flexible photovoltaics and all perovskite or Si-perovskite multijunction solar cells because of their low-temperature processability. Nevertheless, the state-of-the-art efficiencies of p-i-n structured PVSCs suffer from non-ideal interfacial recombination and charge-extraction losses. To address these challenges, we employed a large alkylammonium interlayer (LAI) to reduce the energy loss occurred between transport layers and perovskite. The use of LAIs, in contrast with the reported bottom or top surface passivation strategies, can simultaneously suppress the non-radiative energy losses at both top and bottom interfaces of perovskite. As a result, the reduced surface recombination velocity (SRV) and trap state density (N_t) enable a substantially improved photovoltage from 1.12 to 1.21 V for the PVSCs with an optical band gap (E_g) of 1.59 eV, leading to a champion power conversion efficiency (PCE) over 22%, which is among the highest efficiencies reported for inverted PVSCs.

Context & Scale

The issue of large photovoltage loss is a critical challenge for the development of p-i-n structured PVSCs, which bear great potential for flexible and multijunction solar cells. Here, we report an approach to modulate the defects and interfaces through LAIs for high-performance p-i-n structured PVSCs. By introducing the LAIs between the hole transport layer and perovskite layer in PVSCs, the non-radiative energy losses at both top and bottom interfaces of perovskite were simultaneously suppressed. More importantly, the LAIs have also inhibited the phase segregation on the perovskite surface, enabling a homogeneous surface properties. As a result, a high open-circuit voltage of 1.21 V and a champion PCE of over 22% were achieved, which is among the highest efficiencies reported for p-i-n structured PVSCs. Our work provides an effective strategy to control the optoelectronic properties of perovskite film.

INTRODUCTION

Organic-inorganic PVSCs have emerged rapidly as a promising photovoltaic technology because of the combined excellent optoelectronic properties of perovskite materials, low cost, light weight, and solution processability.[1–5] Recently, a certified PCE of 25.2% has been reported, which can rival that of the prevailing inorganic counterparts including polycrystalline silicon (p-Si), copper indium gallium selenide (CIGS), and cadmium telluride (CdTe). However, most of the PVSCs with certified efficiency over 23% are exclusively based on regular (n-i-p) device configuration with mesoporous TiO_2 scaffold as the electron-transporting layer (ETL), whereas they require high sintering temperatures up to 500°C.[6–8]

By comparison, inverted (i.e., based on p-i-n configuration) PVSCs possess low-temperature processability (~100°C), rendering them more compatible with roll-to-roll processing technique for flexible photovoltaics (PVs).[2,9–11] Moreover, almost all of the perovskite-based tandem solar cells including perovskite-perovskite, Si-perovskite, and CIGS-perovskite multijunction PVs are based on the p-i-n structure.[12–14] Inverted PVSCs have also exhibited impressive operational stability, benefiting from the non-doped hole-transporting layer (HTL) (e.g., poly(triarylamine) (PTAA) or inorganic NiO_x) and highly crystalline perovskite film.[15] Recently, Bai et al. reported a planar inverted PVSCs with long-term operational stability by using ionic liquid additives, which demonstrated less than 5% PCE loss after 1,800 h continuous one-sun light illumination at 70°C–75°C.[3] Then Yang et al. also presented planar inverted PVSCs with wide band gap lead oxysalts to stabilize the perovskite layer and resulted in photovoltaic devices with 97% of its original efficiency retained after being held at the maximum power point (MPP) for 1,200 h under continuous simulated AM 1.5G illumination and 65°C heating.[2]

It has been reported that the maximum photovoltage of a solar cell is dependent on the quasi-Fermi level splitting (QFLS) in the absorber, which is determined by the photo-generated carrier density and intrinsic carrier density in the dark.[16] Non-radiative carrier recombination impairs charge density buildup and diminishes the QFLS, thus leading to a reduced photovoltage of PVSC. Electronic trap states induced by crystallographic defects, including point defects and higher dimensional defects (such as grain boundaries [GBs]), are also the sources for non-radiative charge carrier recombination. In this regard, numerous studies have pointed out the critical roles of point defect- and GB-passivation toward achieving high-efficiency PVSCs.[17–19]

Likewise, interfacial energy loss is also a critical factor that limits the PVSC's photovoltage. Ginger et al. showed that the non-radiative recombination at interfaces has a dominant impact on device performance, and Neher et al. demonstrated that the QFLS in an inverted PVSC can be improved from 1.121 to 1.173 eV via interface passivation.[20] However, compared with the numerous strategies used for passivating the top surface of perovskite film, there are rarely any results mentioned about reducing energy loss from the bottom side. Although the ultrathin polymer films have been successfully utilized to suppress the non-radiative recombination at the bottom interface, these insulating films might hinder charge extraction from the perovskite to the charge-transporting layer (CTL).[21]

In this work, we introduce a LAI between HTL and perovskite for achieving high-efficiency inverted PVSCs. The trap density and interfacial recombination were both effectively suppressed upon employing LAIs, as evidenced by photothermal deflection spectroscopy (PDS), steady-state photoluminescence (PL), and time-resolved PL (TRPL) characterizations. Surprisingly, the bottom LAIs were found not only to act as a passivation layer but also effectively guide the growth process of perovskite films, resulting in higher crystallinity. More importantly, the LAIs have also inhibited the phase segregation on the perovskite top surface, ensuring a more homogeneous interface between the perovskite and ETL. On the basis of these synergistic effects, we achieved a champion PCE of 22.31% with a dramatically increased open-circuit voltage (V_{OC}) from 1.12 to 1.21 V, which is among the highest values reported for inverted PVSCs.

RESULTS AND DISCUSSION

As illustrated in Figure 1A, three kinds of large alkylammonium iodides, phenylethylammonium iodide (PEAI), 1,4-butanediammonium iodide (BDAI), and guanidinium iodide (GAI) were investigated in this work. They were dissolved in dimethylformamide (DMF) and spun onto the PTAA layer. Subsequentially, a mixed cations perovskite based on formamidinium (FA)/methylammonium (MA)/cesium (Cs) was deposited on the PTAA with LAIs. During this process, the large alkylammonium cations interlayer might participate in the interaction with the perovskite precursors in the mixed solvent of DMF and dimethyl sulfoxide (DMSO). As a result, these large alkylammoniums could be incorporated into an interlayer between the PTAA and upper three-dimensional (3D) perovskite because of the high concentration of large alkylammonium cations at the interface.

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https://doi.org/10.1016/j.joule.2020.04.001

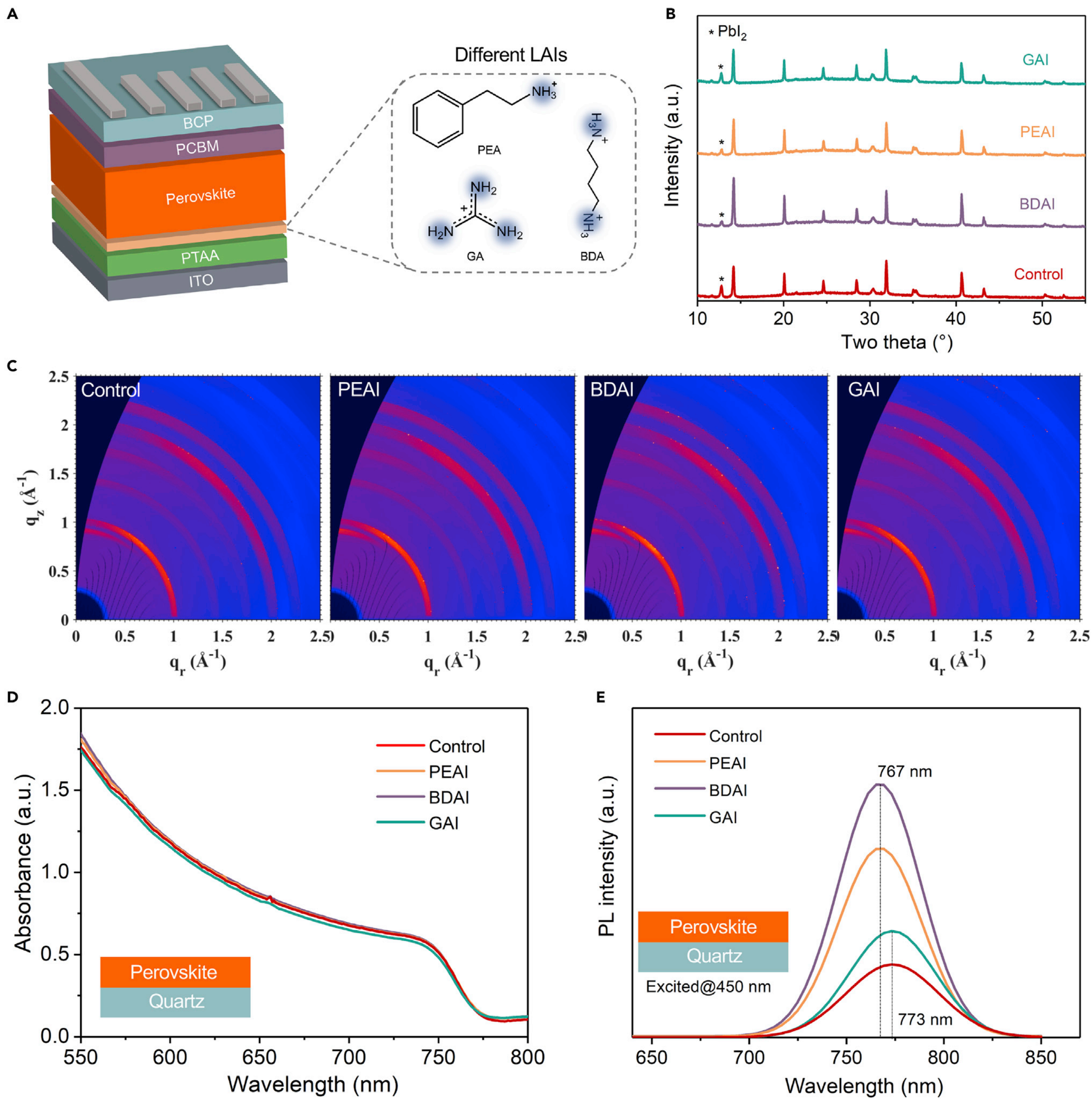

Figure 1. XRD Pattern, GIWAXS, UV-Vis Absorption and PL Spectra of Perovskite Films

(A) Schematic of inverted PVSC structure and the chemical structure of LAIs.

(B) XRD pattern of control film and perovskite films prepared with different LAIs.

(C) GIWAXS patterns with an incident angle of 0.3° for control film and perovskite films prepared with different LAIs.

(D) UV-vis absorption spectra of control film and perovskite films with different LAIs. All the samples were prepared on quartz substrate.

(E) Steady-state PL spectra of control film and perovskite films with different LAIs. All the samples were prepared on quartz substrate.

X-ray diffraction (XRD) measurements were conducted for perovskite films with and without the LAIs to study their crystalline structure, as shown in Figure 1B. All of the as-prepared perovskite films, in spite of different LAIs, exhibited similar perovskite characteristics XRD patterns, where the dominant peaks at 14.16°, 20.10°, and 31.91° were assigned to the FA/MA/Cs mixed-ion perovskite crystal plane diffraction of (001), (011), and (012), respectively.[22] This suggests that the LAIs did not change the lattice structure and induce new phase in perovskite layer. Interestingly, the samples prepared on PEAI, BDAI, and GAI interlayers exhibited stronger peak intensity of (001) than that of control film (without the LAIs). Furthermore, the full width at half maximum (FWHM) of diffraction peaks of the perovskite films with and without LAIs was summarized in Table S1. According to the Scherrer equation, the grain sizes were calculated to be 53.64, 55.77, 56.70, and 54.79 nm for the control film and films with PEAI, BDAI, and GAI as LAI, respectively. The stronger XRD peak intensity and slightly larger grain size for LAI-based perovskite films indicated that the underlying LAIs would change the crystal growth behavior and improve the crystallinity of the upper perovskite to some extent. Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements with an incident angle of 0.3° were also conducted to obtain more information about the crystal structure of the perovskite layer (Figure 1C). Compared with the control sample, no new diffraction pattern appeared for all perovskite films with LAIs, which is consistent with XRD result. This might be because the anticipated interlayer incorporated with LAIs is too thin to be detected by both XRD and GIWAXS measurements in this work. However, the diffraction rings of films with LAIs were slightly brighter than that of the control film, validating the effect of LAIs on improving crystallinity of perovskite films.

To provide more insights about where the LAIs end up, we further conducted GIWAXS measurements with a large incident angle of 1° (Figures S1 and S2) and the steady-state photoluminescence spectra (PL) excited from the quartz (back) and perovskite (front) sides (Figure S3). Compared with the GIWAXS with an incident angle of 0.3°, the indium tin oxide (ITO) patterns (q = 1.52 and 2.14 Å^{-1}) were clearly observed in the GIWAXS with the incident angle of 1°, which indicated that the crystal information of the whole perovskite films was obtained. As a results, there were still no new phases observed in the perovskite films with LAIs. However, from the PL spectra excited from two sides, the weak peak at around 520 nm appeared for the perovskite films with PEAI and BDAI when the films were excited from the back side. This high-energy PL peak might be attributed to the relatively larger band gaps (E_g) of the thin low-dimensional perovskite layer formed at the bottom of the 3D perovskite layer,[23,24] as a result of the reaction between LAIs and the upper 3D perovskite layer. It needs to be pointed out that, the sequential spin-coating process of the perovskite on top of the LAIs is a very complicated and dynamic process, which involves the perovskite formation at the interface; ink formulation; and the change of structure, morphology, and defects of the perovskite. Based on the above considerations and the limitations of characterization technologies to directly examine this bottom interface after the deposition of perovskite layer, it is difficult to conduct simulations or direct chemical analysis to acquire a comprehensive understanding for the LAI's status after spin-coating the upper perovskite layers.

Next, the light-harvesting ability of perovskites were evaluated via ultraviolet-visible (UV-vis) absorption spectra to assess whether the PEAI, BDAI, and GAI-based LAIs would influence the optoelectronic properties of perovskite (Figure 1D). The results indicated that there was minimal impact of LAIs on the absorption and E_g of perovskite films given that all the spectra exhibited a similar profile and absorption band edge. The steady-state PL was then measured to study the radiative recombination behaviors in the perovskite films prepared on quartz. As shown in Figure 1E, the PL intensity of perovskite films based on BDAI, PEAI, and GAI interlayers were about 3.5, 2.6, and 1.5 times higher than that of the control film, respectively, strongly indicating that the non-radiative recombination in the films with LAIs was significantly

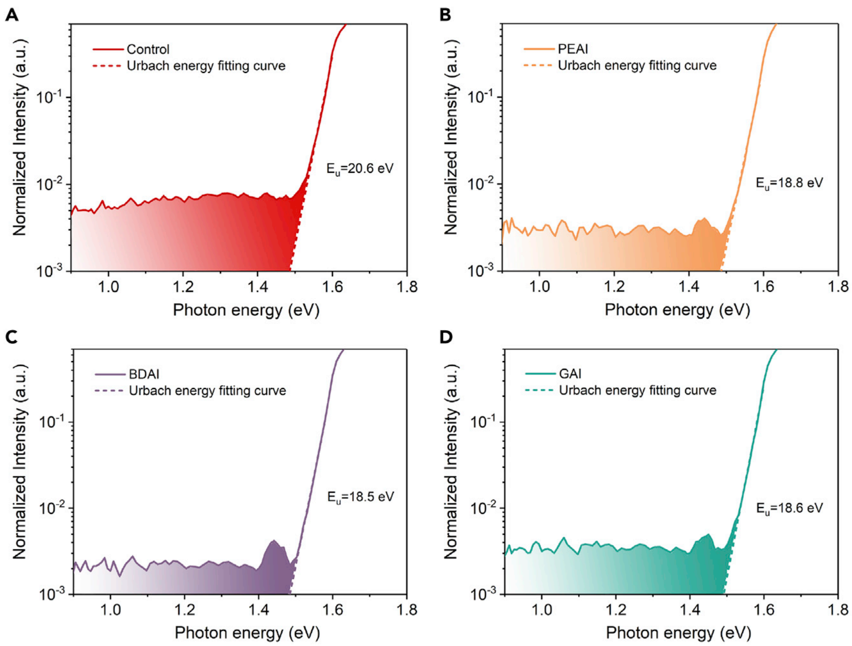

Figure 2. PDS Spectra of Perovskite Films with or without LAIs

(A–D) PDS spectra of perovskite films prepared on quartz substrates: control (A), with PEAI (B), with BDAI (C), and with GAI (D). The colored regions under curves indicate the trap states in the perovskite films.

mitigated. Additionally, a slightly blue-shift of PL peak from 773 nm for the control film to 767 nm for the samples with PEAI and BDAI interlayer was observed, which could be ascribed to the LAI passivating effect on the shallow defects that laid above or below the valence band maximum (VBM) or conduction band minimum (CBM) of perovskite. The similar PL blue-shift was also reported earlier for the Lewis-base passivated defects of perovskite.[25]

The PDS, which is highly sensitive to the sub-band-gap absorption, was employed to quantitatively reveal the passivation effect of LAIs.[26,27] The PDS spectra of perovskite films with and without LAIs were plotted in Figure 2. The band gap of the perovskite was calculated to be 1.59 eV (see Figure S4; Note S1 in Supplemental Information) and the Urbach energy (E_u) was first extracted from the normalized PDS spectra to inspect the band edge disorder of perovskite films. The E_u was calculated to be 20.6, 18.6, 18.8, and 18.5 meV for the control film and the films based on GAI, PEAI, and BDAI, respectively, suggesting that LAIs have a similar effect on reducing the band edge disorder in perovskite films. Moreover, the sub-band-gap absorption of perovskite films with LAIs was found to be significantly lower than that of the control film over the whole range (0.9–1.5 eV). It is noteworthy that there were peaks in the PDS spectra of perovskite films with LAIs in the range of 1.4–1.5 eV, which might have originated from the shallow trap states above/below VBM/CBM. According to the optical sum rule, the N_t of the perovskite films can be quantitatively calculated by the following equation:[26]

$$N_t = \frac{cnm^*}{\pi h e^2} \int \alpha_{sub}(E) dE \qquad \text{(Equation 1)}$$

where c is the speed of light, n is the refractive index, m^* is the effective mass of electron, h is the Planck constant, e is the elementary charge, and E is the photon energy. α_{sub} represents the subtracted sub-gap absorption by Urbach absorption

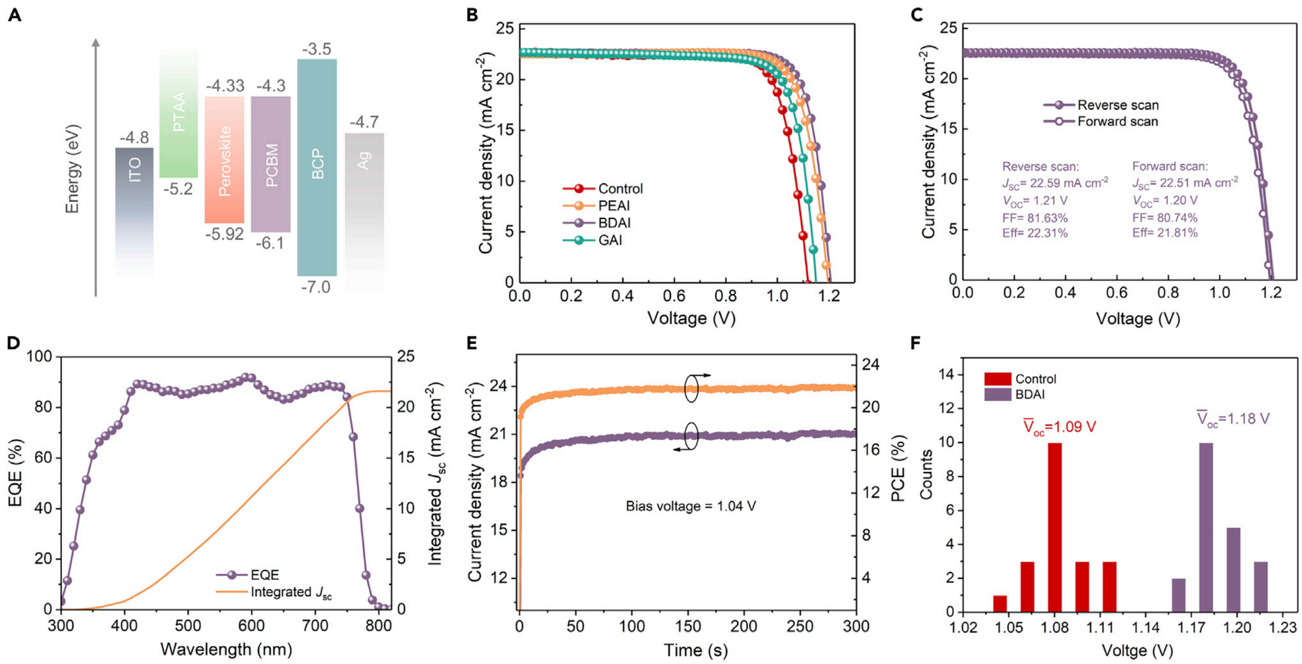

Figure 3. Photovoltaic Performance Characteristics

(A) Energy-level alignment of inverted PVSC with a device structure of ITO/PTAA/perovskite/PCBM/BCP/Ag.

(B) *J-V* curves of control PVSC and PVSCs based on different LAIs.

(C and D) *J-V* curve under reverse and forward scans (C) and EQE spectrum (D) of champion device.

(E) The SPO of champion cell based on BDAI was tracked at the MPP under continuous AM 1.5G illumination.

(F) Histograms of V_{OC} distribution of control PVSC and PVSC based on BDAI. The data of each histogram were extracted from 20 individual cells.

($\alpha_{sub}(E) = \alpha(E) - \alpha_u(E)$, where $\alpha_u(E) = \alpha_0 \exp\left(\frac{h\upsilon - E_g}{E_u}\right)$, E_g is the band gap, all the PDS correlated values are extracted from the PDS spectra as provided in Figure S5). N_t was calculated to be 1.18 × 10^{16} cm^{-3} for the control film and it remarkably decreased to 6.2 × 10^{15}, 5.56 × 10^{15}, and 4.11 × 10^{15} cm^{-3} for films with LAIs based on GAI, PEAI, and BDAI, respectively. Considering the similar E_u of the perovskite films with LAIs, the main reason for the enhanced PL intensity of BDAI-based perovskite film should be attributed to the lowered trap densities.

Given the impressive structural and optoelectronic properties of perovskite films with LAIs, inverted PVSCs were fabricated by one-step method with the device structure of ITO glass/PTAA/perovskite/phenyl-C61-butyric acid methyl ester (PCBM)/bathocuproine (BCP)/Ag. The corresponding schematic of energy-level alignment is displayed in Figure 3A and the energy-levels of control film and the films with GAI, PEAI, and BDAI as LAI were calculated from ultraviolet photoelectron spectra (UPS) and compared in Figure S6. As shown in Figure 3B, the control device yielded a low V_{OC} of 1.12 V, leading to a PCE of 20.09% under 1 sun illumination (AM 1.5G, 100 mW cm^{-2}). When LAIs were introduced between the PTAA/perovskite layer, the device performances were significantly enhanced (Figure S7 and Table S2) and a champion PCE of 22.31% along with a V_{OC} of 1.21 V was achieved for BDAI-based devices, which showed virtually negligible hysteresis (Figure 3C). It has been reported that (two-dimensional) 2D-3D perovskite based on PEAI, BDAI, and GAI possessed Ruddlesden-Popper (R-P) phase, Dion-Jacob (D-J) phase, and alternating-cation-interlayer (ACI) phase, respectively.[28] The best effect of BDAI layer on performance improvement could be due to that the inorganic $[PbX_6]^{4-}$ layers were exposed to the outside in the D-J phase, which would have better lattice match with upper 3D perovskite compared with R-P and ACI phases.

It is noteworthy that the V_{OC} loss, $V_{OC,\ loss} = \frac{E_g}{e} - V_{OC}$, was merely 0.38V for the champion device which was among the lowest values for the inverted PVSCs. The external quantum efficiency (EQE) was plotted in Figure 3D and the integrated short-circuit current density (J_{SC}) from EQE was 21.60 mA cm^{-2}, which is in good agreement with the value extracted from current density-voltage (*J-V*) curve. The stabilized power output (SPO) of device with LAI based on BDAI was tracked at MPP under continuous 1 sun illumination, showing a stabilized efficiency of 21.90% (Figure 3E). The gradually increased efficiency and current density at the beginning could be attributed to the slight light-soaking effect of our device, which was also observed in many reported high-efficiency and stable PVSCs.[7,29,30] In addition, good reproducibility was also achieved as the mean value of V_{OC} was remarkably increased from 1.09 to 1.18 V after utilizing BDAI as LAI (Figure 3F, statistics from 20 individual cells). The key photovoltaic parameters for the devices with different LAIs are summarized in Figure S8 and Table S3 and compared with the reported high-efficiency inverted PVSCs (Table S4).

It is widely accepted that the long alkyl chains in large organic cations will usually cause poor charge extraction because of their inferior conductivity, which will deteriorate both fill factor (FF) and J_{SC}.[31,32] However, in our case, FF and J_{SC} were maintained for PVSCs with LAIs, indicating that these LAIs might only partially cover PTAA surface, therefore, did not impede the charge collection. The trap state densities of perovskite films and the V_{OC} of PVSCs with and without LAIs were provided in Figure S9, presenting a negative relevance between N_t and V_{OC}. This result manifested that the reduction of $V_{OC,\ loss}$ was benefited by reducing trap density.

As mentioned above, the interfacial recombination has dominant impact on device performance, thus the effects of LAIs on the surface properties of upper perovskite layer were investigated to understand the reason causing the V_{OC} enhancement. TRPL with an excitation laser of 485 nm was first carried out to explore the dynamic recombination behaviors in perovskite films prepared on quartz and PTAA-coated quartz (Figure S10), respectively. The decay curves were fitted to a bi-exponential equation and the results were provided in Table S5. The average lifetime can be calculated with the equation of $\tau_{avg} = (A_1\tau_1^2 + A_2\tau_2^2)/(A_1\tau_1 + A_2\tau_2)$, where the A_1 and A_2 is the ratio of the τ_1 and τ_2, respectively. For the films deposited on quartz, the τ_{avg} was calculated to be 684.7 ns for control film whereas it increased to 786.9, 866.4, and 1,038.7 ns for films with LAIs based on GAI, PEAI, and BDAI, respectively (see Supplemental Information). This prolonged τ_{avg} should be attributed to the suppressed non-radiative recombination because of the improved crystallinity of perovskite, which is consistent with the enhanced PL intensity as mentioned before.[33] On the other hand, the PL lifetime of control film deposited on PTAA-coated quartz substantially decreased to 71.4 ns, resulting from either rapid extraction of charge carriers by PTAA or enhanced non-radiative recombination at PTAA/perovskite interface (Figure S10 and Table S6). Nevertheless, a notable recovery of lifetime was found for all the samples with LAIs. It should be pointed out that this recovery was not due to the weakened charge extraction by LAIs but originated from the mitigated interfacial recombination, which was demonstrated by the *J-V* measurements.

To better understand the surface recombination dynamics, (SRV) was calculated, which was independent of the quality of perovskite bulk. The τ_{avg} of PL decay can be described as an equilibrium among the bulky decay (τ_{bulky}), the diffusion time to surface ($\frac{1}{D}\left(\frac{d}{\pi}\right)^2$) and SRV (more details in Note S2; Supplemental Information):[34,35]

$$\tau_{avg}^{-1} = \tau_{bulky}^{-1} + \left(\frac{d}{SRV} + \frac{4}{D}\left(\frac{d}{\pi}\right)^2\right)^{-1} \quad \text{(Equation 2)}$$

where d is the film thickness and D is the diffusion coefficient; D can be estimated from mobility (μ) with Einstein relationship ($D = \mu k_B T/e$), where k_B is the Boltzmann constant, T is the temperature and e is the elementary charge. Here, the average PL lifetime of films prepared on quartz was regarded as τ_{bulky}. Using d = 500 nm (see Figure S11) and μ = 30 $cm^2\ V^{-1}\ s^{-1}$,[36] the SRVs of the films spin-coated on PTAA were calculated as 637.7, 407.2, 201.1, and 149.4 cm s^{-1} for control film, films with LAIs based on GAI, PEAI, and BDAI, respectively. These findings indicate that the LAIs not only suppressed the interfacial non-radiative recombination induced by the surface defects or traps (Figure S12) similar to that previously reported by inserting an insulating polymer interlayer between CTL and perovskite layer,[20] but also increased the charge selectivity and extraction from perovskite to HTL.[37,38] It is noteworthy that the perovskite films with and without LAIs did not exhibit significant structural modifications (Figure 1C), and thus these SRV results manifested that the improvements in performance of PVSCs with LAIs were mainly attributed to the direct interfacial defect passivation.

To test the possible universal applicability of LAIs on other HTLs, the devices with the configuration of: ITO/NiO_x/BDAI/perovskite/PCBM/BCP/Ag and ITO/PEDOT:PSS/BDAI/perovskite/PCBM/BCP/Ag were fabricated (Figure S13). Interestingly, it is found that the V_{OC} and FF both substantially increased for the devices on the basis of PEDOT:PSS as the HTL, whereas there was almost no improvement for the devices using NiO_x as HTL. Considering that the improvements in PVSCs performance were mainly because of the direct interfacial defect passivation, the different performance enhancement on these HTLs could have originated from the different passivation effect, such as the different passivation areas between LAIs and perovskite films. To verify this hypothesis, we conducted the atomic force microscopy (AFM) scanning for PTAA, PEDOT:PSS, and NiO_x films (Figure S14). The roughness of PTAA and PEDOT:PSS was merely 2.0 and 1.5 nm, respectively, whereas it was as high as 8.7 nm for NiO_x. We therefore speculated that the spin-coated LAIs might tend to aggregate at the grain boundaries between the NiO_x particles on the rough surface, whereas they could be more uniformly dispersed on the top of PTAA and PEDOT:PSS films, as illustrated in Figure S15. In this case, the passivated area for NiO_x/perovskite interface would be significantly reduced compared with that for the PTAA or PEDOT:PSS/perovskite interface. Consequently, the passivation effect for NiO_x/perovskite interface should be very limited, and thus there was no obvious improvement in the device performance.

Then, the top-view scanning electron microscopy (SEM) and confocal PL mapping measurements were carried out to investigate the morphologies and optoelectronic properties of the top surface of perovskite (Figure 4). For the PL mapping, considering the film thickness (500 nm, see Figure S11) and the incident depth of the control film (108 nm) and films with GAI (108 nm), PEAI (103 nm) and BDAI (99 nm) as LAIs (calculation details of incident depth are provided in Note S3 in Supplemental Information), the PL signals should be mainly emitted from the top surface of perovskite layer.[39] Although the SEM images showed no obvious changes in morphology and similar crystal size ranging from tens to few hundred nanometers (nm) for all perovskite films (Figure 4A), the confocal PL mapping uncovered interesting PL peak

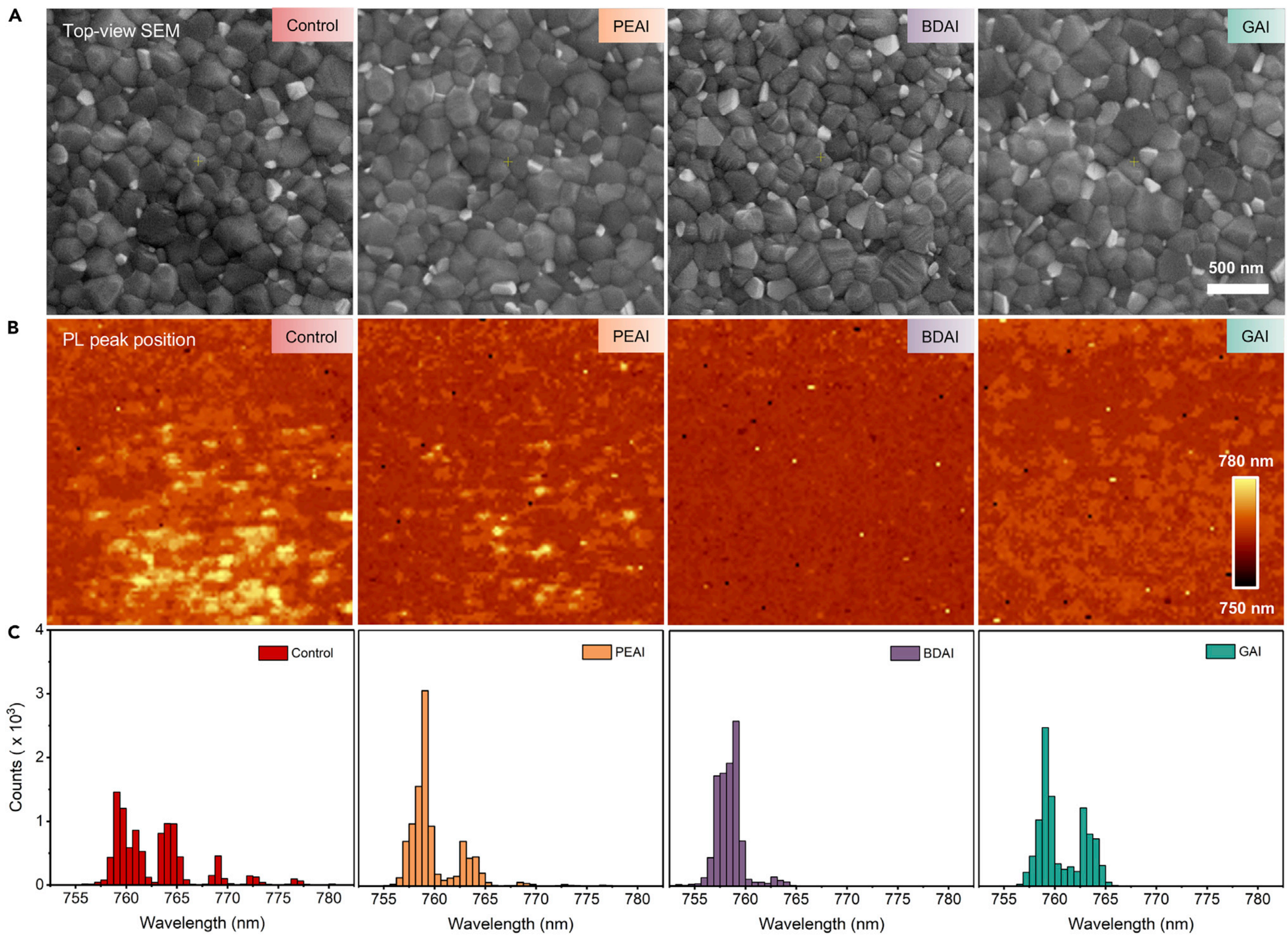

Figure 4. SEM and PL Mapping of Perovskite Films

(A) Top-view SEM image of perovskite films prepared with or without LAIs.

(B) The mapping of PL peak position for perovskite films prepared with or without LAIs (region size: 20 × 20 μm).

(C) Histograms of PL peak position (extracted from PL mapping data) of perovskite films prepared with or without LAIs.

distribution on the perovskite films (Figures 4B and 4C). The PL peak histogram of the control film showed a broad distribution from 757 to 777 nm. However, the distributions of PL peaks were significantly narrowed for perovskite films on the LAIs, suggesting that the surface phase segregation was apparently suppressed (Figure S16).

This phenomenon indicated that the film prepared on LAIs should have more uniform top surface properties, which could potentially reduce the energy loss induced by energy transfer between the domains with different E_gs and various interfacial energy alignment.[40,41] On the other hand, the PL intensity mapping of perovskite films exhibited the similar distribution behavior except for the sample prepared on BDAI interlayer, which showed a slightly narrower intensity distribution (Figure S17). These observations pointed out that the bottom-interfacial recombination behavior and top surface properties could be simultaneously controlled via perovskite/PTAA interfacial modulation. To have a deeper understanding of this intriguing effect of LAIs on the top surface properties of perovskite film requires further studies, however, it is beyond the topic of this work and will be reported in due course.

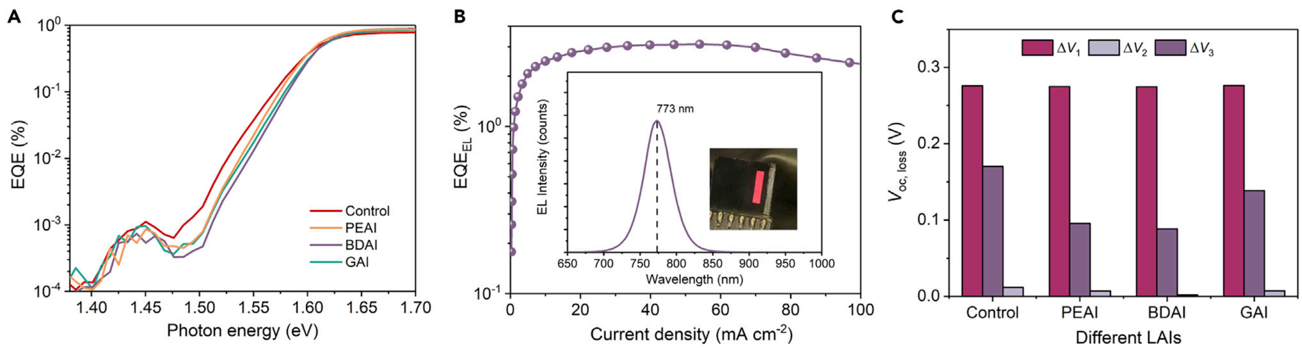

Figure 5. Photovoltage Loss Analysis and Calculation of PVSCs

(A) Highly sensitive EQE of control PVSC and PVSCs based on different LAIs.

(B) EQE_{EL} and EL spectrum (inset) of PVSCs based on BDAI as LAI.

(C) V_{OC} loss analysis of control PVSC and PVSCs based on different LAIs.

To quantitatively estimate the effect of LAIs on V_{OC} loss, the detailed balance theory[42–44] was employed to calculate the V_{OC} loss, which can be attributed to three factors (detailed calculation of V_{OC} loss is described in Note S4 in Supplemental Information):

$$
\begin{aligned}
e\Delta V &= E_g - qV_{OC} \\
&= \left(E_g - eV_{OC}^{SQ}\right) + \left(eV_{OC}^{SQ} - eV_{OC}^{rad}\right) + \left(eV_{OC}^{rad} - eV_{OC}\right) \\
&= \left(E_g - eV_{OC}^{SQ} + e\Delta V_{OC}^{SQ}\right) + \Delta eV_{OC}^{rad} + \Delta eV_{OC}^{non-rad} \\
&= \left(e\Delta V_1 + \Delta V_2 + \Delta V_3\right)
\end{aligned}
\qquad \text{(Equation 3)}
$$

where ΔV is the total calculated voltage loss, V_{OC}^{SQ} is the Shockley-Queisser limit of V_{OC}, V_{OC}^{rad} is the radiative limited of V_{OC} in PVSCs, ΔV_{OC}^{SQ} is the V_{OC} loss due to the non-ideal EQE above E_g, ΔV_{OC}^{rad} is the V_{OC} loss originated from the sub-band-gap radiative recombination and ΔV_{OC}^{rad} is the V_{OC} loss due to the non-radiative recombination.

The term of ΔV_1 is originated from the black-body radiation and the non-ideal EQE above E_g. The PVSCs with and without LAIs showed similar ΔV_1 of around 270 mV, as summarized in Table S7. ΔV_2 is ascribed to the energy loss associated with extra thermal radiation in a solar cell, where EQE of PVSCs extends into the region below E_g therefore results in more black-body radiation. This term can be calculated from the highly sensitive EQE of the PVSCs (Figure 5A).[45,46] The calculated ΔV_2 of PVSCs without and with LAIs are 11.72, 7.42, 7.21, and 1.80 mV, respectively (Table S7). These small ΔV_2 could be ascribed to the steep absorption edge of our perovskite films, which is consistent with the previous report.[47,48]

Additionally, it is noteworthy that the highly sensitive EQE spectra of PVSCs all have a peak in the range of 1.4–1.5 eV, which are located at the same range that the peaks were also observed in the PDS spectra, hence we speculated that these EQE peaks presented the photocurrent contribution from shallow trap states. ΔV_3 is the V_{OC} loss from the non-radiative recombination, which can be deduced with the equation listed below[43,44]:

$$\Delta V_3 = \frac{k_B T}{q} \ln(EQE_{EL}) \qquad \text{(Equation 4)}$$

where EQE_{EL} is the EQE of electroluminescence (EL). The EQE_{EL} was measured by operating the PVSCs as light-emitting diodes (LEDs), as shown in Figures 5B and S18. The EQE_{EL} as high as 3.11% can be obtained with the EL spectra centered at 773 nm for the device with BDAI. The EQE_{EL} of 2.89% at 1 sun equivalent current injection of champion solar cell was used in Equation 4, acquiring a ΔV_3 of 88.60 mV, which was among the lowest reported values for PVSCs. In comparison, the ΔV_3 of the control device was up to 170.31 mV. The values of ΔV_1, ΔV_2, and ΔV_3 of the PVSCs without and with LAIs are displayed in Figure 5C and summarized in Table S7. The calculated V_{OC} matched well with our measured V_{OC}. This result indicated that the ΔV_3 dominated the voltage loss of PVSCs in our work, further supporting the effect of LAIs on reducing the non-radiative recombination through the modulation of defects and interfaces of perovskite.

The long-term stability of encapsulated PVSCs was finally investigated in ambient atmosphere with a relative humidity of 20% and temperature of 25°C (Figure S19). The shelf stability of the devices with LAIs were more stable than that of the control device, exhibiting a trend of BDAI > PEAI > GAI > Control. The PCE of the device with BDAI could retain 87% of the initial value after 850 h, whereas it significantly dropped to 60% of the initial efficiency for the control device. It has been reported that the defect sites not only militate against a high photovoltage, but also shorten the lifetime of perovskite devices as they present the vulnerable sites for initiating degradation by extrinsic environmental stimuli (i.e., moisture, oxygen, etc.). Therefore, the enhanced stability should be derived from the less trap states density and surface passivation. Moreover, the SPO of the PVSC based on BDAI was tracked at MPP under a white LED equal to AM 1.5G illumination in a N_2-filled glovebox. The PCE retained 98.6% of the initial value after 10 h illumination (Figure S20), which proved a good long-term photostability of the PVSC based on BDAI.

Conclusion

In summary, an effective strategy was demonstrated by employing LAIs deposited between PTAA and perovskite layer to control the growth of perovskite film and suppress the non-radiative recombination. Although the LAI does not affect the crystal structure and morphology of upper perovskite layer, it improves the optoelectronic properties of perovskite film, leading to stronger PL intensity and more uniform PL peak distribution on the top surface. Furthermore, the reduced SRV in PTAA/perovskite with LAIs indicates the suppressed non-radiative recombination at interfaces and enhanced charge selectivity from perovskite to HTL. With these combined benefits, a champion PCE of 22.31% was achieved with a significantly enhanced V_{OC} of 1.21 V for the inverted PVSC based on BDAI. Our work provides significant insights for controlling perovskite film properties for further improving the efficiency of perovskite photovoltaics and optoelectronic devices.

EXPERIMENTAL PROCEDURES

Materials

Cesium iodide (CsI), formamidinium iodide (FAI), and methylammonium bromide (MABr) were purchased from Dysol (Australia). Lead iodide (PbI_2, 99.9985%) and lead bromide ($PbBr_2$, 99.9985%) were purchased from TCI (Japan). Phenyl-C61-butyric acid methyl ester (PCBM) was purchased from ADS. GAI, PEAI, BDAI, PTAA, and BCP, 99.9% were purchased from Xi'an Polymer Light Technology Corp. (China). The liquid reagents, including N, N-dimethylformamide (DMF, 99.8%), DMSO, 99.7%, isopropanol (IPA, 99.5%), and chlorobenzene (CB, purity of 99.8%) were purchased from J&K (China) and used as received. Toluene

(99.7%) was purchased from Tian Hang Technology Co., Ltd. Silver was purchased from commercial sources with high purity.

Device Fabrication of PVCs

The pre-patterned ITO, 15 Ω sq^{-1}) glass substrates were sequentially cleaned by sonication with detergent, DI water, acetone and IPA for 20 min, respectively. Then, the cleaned ITO glass substrates were dried in a hot oven at a temperature of 100°C. Before use, the ITO substrates were treated with oxygen plasma for 15 min and transferred into N_2-filled glovebox. Then, 2 mg mL^{-1} PTAA in toluene was firstly spun onto ITO substrates at 5,000 rpm (with a ramping rate of 2,500 rpm s^{-1}) for 30 s, and were subsequently annealed at 100°C for 15 min. When the substrate was cooled for 5 min, the alkylammonium iodide solution (GAI, PEAI or BDAI with different concentration in DMF) was dripped onto PTAA layer and spin-coated at 4,000 rpm for 30 s, and then annealed at 100°C for 5 min. The perovskite precursor solution was prepared by mixing CsI, FAI, MABr, PbI_2, and $PbBr_2$ in DMF: DMSO (5: 1/v: v) with a chemical formula of $(CsPbI_3)_{0.05}$(-$FA_{0.85}MA_{0.15}Pb(I_{0.85}Br_{0.15})_3)_{0.95}$. For the control device, 50 μl DMF was first used to treat PTAA layer to improve its wetting ability before the deposition of perovskite. Then, 60 μl of perovskite precursor solution was spin-coated on PTAA layer at 1,000 rpm (with a ramping rate of 500 rpm s^{-1}) for 10 s and 5,000 rpm (with a ramping rate of 2,000 rpm s^{-1}) for 30 s. During the second step, 110 μl CB as anti-solvent was quickly dripped on the center of perovskite film at 15 s before the end of spin-coating and then annealed at 60°C for 5 min and 100°C for 30 min. PCBM (20 mg mL^{-1} in CB) was spun onto the perovskite layer ata speed of 1,200 rpm (with a ramping rate of 500 rpm s^{-1}) for 30 s. All the spin-coating processes were conducted in a N_2-filled glovebox with the contents of O_2 and H_2O <1 ppm. Finally, 5 nm BCP and 100 nm Ag were thermally evaporated in a high vacuum chamber (<2×10^{-6} Torr) through a metal shadow mask (aperture area: 0.13 cm^2), respectively.

Films and Devices Characterizations

UV-vis absorption spectra were measured by a UV-vis spectrometer (PerkinElmer model Lambda 2S). Thin film XRD characterization was carried out in a D2 Phaser instrument (Bruker) with Cu Kα (λ = 0.154 nm) radiation. The GIWAXS results were recorded at beamline BL14B1 of the Shanghai Synchrotron Radiation Facility (SSRF) by using X-ray with a wavelength of 0.6887 Å. All photoemission studies were carried out in a VG ESCALAB 220i-XL surface analysis system equipped with a He-discharge lamp (hv = 21.22 eV) for UPS investigation. The film thickness of the perovskite films was measured by a DektakXT stylus profiler and the results of the average thickness were calculated from the three repeated samples. Steady-state photoluminescence (PL) and time-resolved photoluminescence spectrum (TRPL) were recorded by FLS980 spectrofluorometer (Edinburgh) with an excitation light of 480 nm and a pulsed excitation laser of 485 nm, respectively. Photoluminescence imaging was carried out on WITec alpha300 M+ confocal microscopy (WITec GmbH). The excitation laser was a diode-pumped solid-state laser (532 nm, cobalt Laser). A diffraction-limit spot size of the 532 nm laser was obtained about 300 nm with a 100× objective (NA = 0.90). The photoluminescence was dispersed by a 600 mm focal length spectrometer with 1,800 and 300 grooves/mm grating (UHTS 600), respectively. The signals were eventually detected by using a charge-coupled-device (CCD) thermoelectrically cooled to −60°C. PDS measurements were carried out with a standard setup consisting of 1 kW Xe arc lamp and a 0.25 m grating monochromator (Oriel) as the tunable light source. The pump beam was modulated at 13 Hz by a mechanical chopper before irradiating on the sample. Perfluorohexane was used as the deflection fluid. A Uniphase HeNe laser was directed parallel to the polymer sample surface as the probe laser. A quadrant cell (United Detector

Technology) was used as the position sensor for monitoring the photothermal deflection signal of the probe beam. The output of the detector was fed into a lock-in amplifier (Stanford Research, Model SR830) for phase-sensitive measurements. All PDS spectra were normalized to the incident power of the pump beam. The current density-voltage (*J–V*) characteristics of photovoltaic devices were measured in a N_2-filled glovebox at room temperature by using a Keithley 2400 sourcemeter under simulated sunlight from a solar simulator (EnliTech, SS-F5, Taiwan). A National Renewable Energy Laboratory calibrated silicon solar cell (with a KG-2 filter) was used to calibrate the AM 1.5G (100 mW cm^{-2}) solar simulator's light intensity. The PVCs were covered with a shading mask with an aperture area of 0.104 cm^2 to ensure the accuracy of current density in *J-V* curves. The *J-V* measurements was conducted with sweep mode with reverse (from 1.23 to −0.02 V) and forward (from −0.02 to 1.23 V) scan with a scan rate of 300 mV s^{-1}. EQEs were carried out by an EQE measurement system (EnliTech, QE-R, Taiwan).

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.joule.2020.04.001.

ACKNOWLEDGMENTS

The work was supported by the New Faculty Start-up Grant of the City University of Hong Kong (9610421 and 7200587), the Office of Naval Research (N00014-17-1-2201), the GRF grant (CityU 21301319), and a Collaborative Research Fund grant (C5037-18G) from the Research Grants Council of Hong Kong, and the Innovation and Technology Support Programme (ITS/497/18F and GHP/021/18SZ). Z.Z. and A.J. thank the beam time and technical support provided by BL14B1 beamline at Shanghai Synchrotron Radiation Facility (SSRF) Laboratory.

AUTHOR CONTRIBUTIONS

S.W., J.Z., and Z.L. contributed equally to this work. Z.Z. and A.J. conceived the ideas and designed the project. Z.Z. and A.J. directed and supervised the research. S.W. fabricated and characterized devices. Z.L. and J.Z. also contributed to device fabrication, characterizations, and helped S.W. analyze data. J.Z. conducted the calculations of SRV and V_{OC} loss. S.C. and S.S. helped to conduct the PDS measurement. M.Q. and X.L. helped to analyze GIWAXS data. D.J.L. and D.Y.L. helped to conduct PL mapping measurement. S.W., J.Z., Z.Z., and A.J. drafted and finalized the manuscript. All the authors revised the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: December 28, 2019
Revised: March 9, 2020
Accepted: April 1, 2020
Published: April 27, 2020

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